

Binary regression with differentially misclassified response and exposure variables

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Misclassification is a long-standing statistical problem in epidemiology. In many real studies, either an exposure or a response variable or both may be misclassified. As such, potential threats to the validity of the analytic results (e.g., estimates of odds ratios) that stem from misclassification are widely discussed in the literature. Much of the discussion has been restricted to the nondifferential case, in which misclassification rates for a particular variable are assumed not to depend on other variables. However, complex differential misclassification patterns are common in practice, as we illustrate here using bacterial vaginosis and Trichomoniasis data from the HIV Epidemiology Research Study (HERS). Therefore, clear illustrations of valid and accessible methods that deal with complex misclassification are still in high demand. We formulate a maximum likelihood (ML) framework that allows flexible modeling of misclassification in both the response and a key binary exposure variable, while adjusting for other covariates via logistic regression. The approach emphasizes the use of internal validation data in order to evaluate the underlying misclassification mechanisms. Data-driven simulations show that the proposed ML analysis outperforms less flexible approaches that fail to appropriately account for complex misclassification patterns. The value and validity of the method are further demonstrated through a comprehensive analysis of the HERS example data. Copyright © 2015 John Wiley & Sons, Ltd.

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1. Introduction

For many epidemiologic studies, reliably estimating effects of exposures on a health outcome is of primary interest. It has been noted in a wide range of study contexts [1–8] that the problem of measurement error may arise because of fallibility in measurement tools, hindering investigators in the effort to report valid estimates of effect. In addition to these references, many general discussions about the impact of measurement error on estimating health effects and approaches to combat it can be found in the literature (e.g., [9–14]).

When referring to measurement error in binary or categorical variables, the term misclassification is commonly used, and it has been sometimes considered as a distinct concept. For example, to assess disease status, epidemiologic studies sometimes use convenient but error-prone diagnostic tests, which may lead to false-positive and false-negative diagnoses. There is rich literature on methods dealing with misclassification in exposure or response variables, often focusing on nondifferential misclassification or invoking sensitivity analyses when errors may be differential (e.g., [15–18]). However, the value of internal validation data for elucidating and adjusting for exposure misclassification that is differential with respect to disease status has been made very clear (e.g., [11, 19–22]). While some publications advocate the use of internal validation data to evaluate a more complex misclassification process, relatively

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few emphasize practical guidance on how to implement the analysis. Our current work seeks to address aspects of the gap between theoretical considerations and real-life applications when one encounters potentially complex patterns of binary variable misclassification.

Here, we consider a motivating example arising in the HIV Epidemiology Research Study (HERS) [23], where both a binary exposure variable X (Trichomoniasis (TRICH) status) and a binary response variable Y (bacterial vaginosis (BV) status) are subject to misclassification. The presence of outcome and/or exposure misclassification complicates regression analyses and can lead to distorted effect estimates (e.g., estimated odds ratios (ORs)). Moreover, the misclassification process with respect to BV in the HERS data has been found to be quite complex, making one doubt both the common assumption of nondifferentiality and typical differential mechanisms in which false-positive and false-negative rates depend only on a single variable. Exploration into the misclassification of BV status further revealed that inappropriately assuming nondifferentiality can lead to corrected OR estimates as biased as or more biased than those arising from 'naïve' analyses that ignore the problem altogether [24].

In the following section, we first introduce the HERS data and discuss analytical questions arising with regard to logistic regression analysis of cross-sectional data involving TRICH and BV status. When misclassification is present in both X and Y , prior discussions in the literature mainly target two-by-two tables or generalized tables [25, 26]. However, in practice, it is very common to seek direct adjustment for covariates via regression modeling, as in our motivating example. Thus, we propose a unified likelihood framework to incorporate the main/internal validation design and assess complex misclassification mechanisms involving both binary variables. We present data-driven simulations mimicking the HERS example to demonstrate the benefits of the main/internal validation study design and corresponding likelihood-based methods, relative to a naïve analysis and an alternative approach that adjusts for misclassification based on overly simplistic assumed mechanisms. We also conduct extensive simulations under various settings to evaluate the performance of the proposed approach.

2. Materials and methods

2.1. Motivation

We consider a cross-sectional analysis of data on BV as the response Y and TRICH status as the exposure X for women in the HERS at the fourth semi-annual visit (between 09/1994 and 10/1996), given misclassification of BV status previously documented at that visit [24]. The primary goal of the analysis is to model the association of BV with TRICH, controlling for other subject-specific characteristics. Existing literature suggests a positive relationship between these two conditions [27], highlighting the motivation for assessing their prevalence and association. Unfortunately, in practice, BV and TRICH may be diagnosed via error-prone tests. In HERS, these consist of a clinical method (hereafter referred to as 'CLIN') for BV assessment based on a modified version of Amsel's criteria [28] and a wet mount testing procedure [29] for detecting TRICH. Although such clinically based methods are convenient and cost saving, they are subject to yielding false positives or negatives. In large-scale observational studies like HERS that seek insight about disease associations, the presence of misclassification in such diagnoses jeopardizes the validity of adjusted OR estimates and corresponding statistical inferences.

A total of 904 women with complete data on BV, TRICH, and other risk factors at the fourth visit were considered. Among them, 61.7% were Black, 67.4% were HIV positive, and 52% were intravenous drug users. The median age at enrollment was 37 years. Using culture testing (the arguable gold standard), 18% of women were diagnosed with TRICH. In contrast, only 7.6% were TRICH positive based on wet mounts. Similarly, 40.3% of women were BV positive via a gold standard laboratory-based ('LAB') method based on Nugent's criteria [30], compared with 24.5% based on the CLIN method. Estimated crude sensitivities for the wet mount and CLIN methods were 37.8% and 51.7%, respectively, while the specificities were 93.9% and 99.0%, indicating that both error-prone methods were highly specific, but not sensitive.

We refer to the binary random variables corresponding to the gold standard TRICH and BV assessments as X and Y , respectively. The error-prone assessments are conversely labeled as X^* and Y^* . In HERS, all four assessments (X , X^* , Y , and Y^*) were obtained for each participant at visit 4. The complete data on X and Y provide an ideal data example to illustrate the performance of the proposed approach, as we can compare the results of a standard logistic regression applied to the gold standard data with those of a naïve analysis based on (X^*, Y^*) and with maximum likelihood (ML) analyses incorporating only a portion of the (X, Y) data into an internal validation sample.

Table I. Logistic regression results for BV status (Y) on 904 women at the fourth HIV Epidemiology Research Study visit.

Variable	$\hat{\beta}$ (SE)	$\widehat{OR}$ (95% CI)
Ideal analysis ^a		
Trichomoniasis (+ versus −)	0.88 (0.19)	2.41 (1.66, 3.50)
Age (years)	−0.04 (0.01)	0.96 (0.94, 0.98)
Race (Black versus others)	0.76 (0.16)	2.15 (1.57, 2.92)
HIV risk cohort (IDU versus sexual)	0.31 (0.15)	1.37 (1.03, 1.83)
HIV status (+ versus −)	0.22 (0.15)	1.25 (0.93, 1.69)
Naïve analysis ^b		
Trichomoniasis (+ versus −)	1.24 (0.27)	3.44 (2.03, 5.84)
Age (years)	−0.05 (0.01)	0.95 (0.93, 0.98)
Race (Black versus others)	0.69 (0.18)	1.99 (1.40, 2.83)
HIV risk cohort (IDU versus sexual)	0.90 (0.17)	2.45 (1.74, 3.45)
HIV status (+ versus −)	−0.31 (0.17)	0.73 (0.52, 1.03)

BV, bacterial vaginosis; IDU, intravenous drug use; SE, standard error.

^aLaboratory-based BV versus culture Trichomoniasis, adjusting for age, race, HIV risk cohort, and HIV status.

^bClinical BV versus wet mount Trichomoniasis, adjusting for age, race, HIV risk cohort, and HIV status.

We first consider a multiple logistic model of interest on all subjects using Y and X as the response and predictor variables, referred to as the ‘ideal analysis’. Preliminary model selection suggested that TRICH status, age, race (Black versus others), HIV risk cohort (*RISKCHRT*: intravenous drug use versus sexual), and HIV status (*HIVPOS*: positive versus negative) are important risk factors for BV, as represented in model (1):

$$\text{logit}[\Pr(Y = 1)] = \beta_0 + \beta_1 X + \beta_2 \text{AGE} + \beta_3 \text{RACE} + \beta_4 \text{RISKCHRT} + \beta_5 \text{HIVPOS} \quad (1)$$

We then fit the same model replacing X and Y with X^* and Y^* , denoted as the ‘naïve analysis’. As seen in Table I, the two analyses differ markedly in magnitudes of the estimated OR for TRICH (2.41 with 95% CI (1.66, 3.50) for ideal vs. 3.44 with 95% CI (2.03, 5.84) for naïve) and HIV risk cohort (1.37 with 95% CI (1.03, 1.83) for ideal vs. 2.45 with 95% CI (1.74, 3.45) for naïve). The estimated ORs for HIV status differ in directionality (1.25 with 95% CI (0.93, 1.69) for ideal vs. 0.73 with 95% CI (0.52, 1.03) for naïve). These discrepancies clearly indicate that without appropriately taking misclassification into account, estimates of effect and corresponding inferences that rely only upon X^* and/or Y^* can be invalid.

2.2. Misclassification patterns and joint likelihood formulation

Consider a cross-sectional study with n subjects. In the absence of misclassification, assume that we seek to fit a standard logistic regression model as follows:

$$\text{logit}[\Pr(Y = 1|X, C_1, C_2 \cdots C_P)] = \beta_0 + \sum_{p=1}^P \beta_p C_p + \beta_{P+1} X \quad (2)$$

where Y is a potentially misclassified binary response variable, C_p ($p = 1, \dots, P$) denotes the p th covariate assumed to be measured accurately, and X stands for a binary predictor of interest that (like Y) is subject to misclassification. Henceforth, we divide the whole study sample into two parts. In the main study sample, instead of X and Y , mismeasured dichotomous exposure status X^* and disease status Y^* are observed. For the purpose of evaluating misclassification patterns, we assume that a completely random internal validation sample of size n_v is selected and gold standard measures of the response and exposure (Y and X) are made on those subjects. It follows that the main study sample size is $n_m = n - n_v$. We note in passing that some authors refer to the whole study sample as the ‘main study’, which differs slightly from our use of the term. If we replace X and Y in Eq. (2) with the error-prone measures X^* and Y^* , estimates of the β ’s will be potentially biased (even in large samples). The magnitudes of the biases impacting these estimates depend on the diagnostic properties of the methods used to classify both X^* and Y^* .

2.2.1. Independent nondifferential misclassification. The assumption of nondifferential misclassification implies the belief that the common diagnostic parameters known as sensitivity (SE) and specificity (SP) are constants that do not vary based upon any subject-specific variables (e.g., [7]). For example, regarding diagnostic properties relating Y^* to Y in the nondifferential case, we define

$$SE_Y = \Pr(Y^* = 1|Y = 1) \text{ and } SP_Y = \Pr(Y^* = 0|Y = 0) \quad (3)$$

Similarly, when characterizing the method classifying X^* , nondifferentiability implies that

$$SE_X = \Pr(X^* = 1|X = 1) \text{ and } SP_X = \Pr(X^* = 0|X = 0) \quad (4)$$

The subscript notation used in (3) and (4) reflects the fact that SE and SP are viewed as constants independent of other information, such as disease status and prognostic factors. In other words, nondifferential misclassification for both variables corresponds to the following assumptions: $\Pr(Y^* = 1|Y = 1) = \Pr(Y^* = 1|Y = 1, X, C)$ and $\Pr(X^* = 1|X = 1) = \Pr(X^* = 1|X = 1, Y, C)$. It follows that $\Pr(Y^*, X^*, Y, X|C) = \Pr(Y^*|Y, X^*, X, C) \Pr(X^*|X, Y, C) \Pr(Y|X, C) \Pr(X|C) = \Pr(Y^*|Y) \Pr(X^*|X) \Pr(Y|X, C) \Pr(X|C)$, indicating the misclassification processes in X and Y are independent. This type of misclassification pattern is hereafter termed ‘independent nondifferential misclassification’.

We set out to find expressions for the observed data likelihood under various misclassification assumptions. Let us first consider further the situation when both X and Y are assumed subject to independent nondifferential misclassification. By the rule of total probability, each independent observation in the main study contributes the following likelihood term:

$$\begin{aligned} \Pr(Y^* = y^*, X^* = x^*|C = c) &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(y^*, x^*, y, x|c) \\ &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(y^*|y) \Pr(x^*|x) \Pr(y|x, c) \Pr(x|c) \end{aligned} \quad (5)$$

The first and second terms in the last line of Eq. (5) represent the SE and SP of Y and X , while the third term reflects the primary model of interest defined in Eq. (2). The last term characterizes the association of X with other covariates C . To facilitate the likelihood specification, we designate a model for $\Pr(X|C)$. While other links can be applied, here, we adopt the familiar logit link and assume that Eq. (6) is correctly specified:

$$\text{logit}[\Pr(X = 1|C_1, C_2 \cdots C_Q)] = \gamma_0 + \sum_{q=1}^Q \gamma_q C_q \quad (6)$$

The overall likelihood for the main study follows as

$$L_m = \prod_{i=1}^{n_m} \left\{ \sum_{y_i=0}^{y_i=1} \sum_{x_i=0}^{x_i=1} \Pr(y_i^*|y_i) \Pr(x_i^*|x_i) \Pr(y_i|x_i, c_{iy}) \Pr(x_i|c_{ix}) \right\} \quad (7)$$

where the subscripts of c in Eq. (7) indicate that the version of the c vector may differ across models as long as c_{ix} is a subset of c_{iy} . The scenario just described assuming nondifferential misclassification of both X and Y is an extension of the classical 2×2 setting [2] to incorporate an arbitrary set of accurately measured covariates (C) into the model associating X with Y . In [2], Barron presented an identity, later termed the ‘matrix method’, to connect the correct and error-prone cell counts (or probabilities) via the bridge of the diagnostic parameters SE_x , SE_y , SP_x , and SP_y . Unless one is to rely upon sensitivity analyses, these parameters are best estimated by supplemental sampling of subjects into an internal validation set. According to the strategy discussed following Eq. (2), we assume that (X, X^*, Y, Y^*) are observed on each

subject in the internal validation sample (although only minor adjustments to the likelihood contributions are required, e.g., if certain subjects have only one variable validated). The likelihood contribution for the internal validation subsample is

$$L_v = \prod_{j=1}^{n_v} \Pr(y_i^* | y_i) \Pr(x_i^* | x_i) \Pr(y_i | x_i, c_{iy}) \Pr(x_i | c_{ix}) \quad (8)$$

and the overall joint likelihood to be maximized is the product of the expressions in (7) and (8).

2.2.2. Independent differential misclassification. In contrast to the nondifferential case, differential misclassification occurs when the misclassification probabilities of one variable depend on the value(s) of the other variable(s). More specifically, regarding classifying via Y^* , we define

$$SE_{Yxc} = \Pr(Y^* = 1 | Y = 1, X = x, C = c)$$

and

$$SP_{Yxc} = \Pr(Y^* = 0 | Y = 0, X = x, C = c) \quad (9)$$

Note that the SE and SP of Y now can be functions of exposure (X) and other covariates (C). For clarification, as long as the SE 's/ SP 's for X and Y depend only on the true values of other variables, we will term it 'differential but independent misclassification'. Equation (9) is the most general representation of SE_Y and SP_Y under such an assumption and implies the possibility of modeling SE_Y and SP_Y parametrically. Similarly, X may or may not be an important factor in characterizing the diagnostic properties for Y and can be left out when deemed not to be. We may define the misclassification process for X analogously, as follows:

$$SE_{Xyc} = \Pr(X^* = 1 | X = 1, Y = y, C = c)$$

and

$$SP_{Xyc} = \Pr(X^* = 0 | X = 0, Y = y, C = c) \quad (10)$$

Each observation in the main study now contributes to the likelihood as follows:

$$\begin{aligned} \Pr(Y^* = y^*, X^* = x^* | C = c) &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(y^*, x^*, y, x | c) \\ &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(y^* | y, x, c) \Pr(x^* | x, y, c) \Pr(y | x, c) \Pr(x | c) \end{aligned} \quad (11)$$

Note that Eq. (11) is a generalization of (7) to incorporate differential misclassification. The first two terms in the last line of (11), characterizing SE and SP for X and Y , need to be specified in order to write out the likelihood. Here, again, we favor logistic regressions for modeling the misclassification processes of X and Y . More specifically, define

$$\text{logit}[\Pr(Y^* = 1 | Y, X, C_1, C_2 \dots C_R)] = \theta_0 + \sum_{r=1}^R \theta_r C_r + \theta_{R+1} Y + \theta_{R+2} X \quad (12)$$

and

$$\text{logit}[\Pr(X^* = 1 | Y, X, C_1, C_2 \dots C_S)] = \delta_0 + \sum_{s=1}^S \delta_s C_s + \delta_{S+1} X + \delta_{S+2} Y \quad (13)$$

Model (12) implies that

$$SE_{Yxc} = \frac{\exp\left(\theta_0 + \sum_{r=1}^R \theta_r C_r + \theta_{R+1} + \theta_{R+2} X\right)}{1 + \exp\left(\theta_0 + \sum_{r=1}^R \theta_r C_r + \theta_{R+1} + \theta_{R+2} X\right)}$$

and

$$SP_{Yxc} = \frac{1}{1 + \exp\left(\theta_0 + \sum_{r=1}^R \theta_r C_r + \theta_{R+2} X\right)}$$

The corresponding definitions of SE_{Xyc} and SP_{Xyc} are determined analogously from model (13). The full likelihood follows as $L = L_m \times L_v$, where

$$L_m = \prod_{i=1}^{n_m} \left\{ \sum_{y_i=0}^{y_i=1} \sum_{x_i=0}^{x_i=1} \Pr(y_i^* | y_i, x_i, \mathbf{c}_{iy*}) \Pr(x_i^* | x_i, y_i, \mathbf{c}_{ix*}) \Pr(y_i | x_i, \mathbf{c}_{iy}) \Pr(x_i | \mathbf{c}_{ix}) \right\} \quad (14)$$

and

$$L_v = \prod_{j=1}^{n_v} \Pr(y_j^* | y_j, x_j, \mathbf{c}_{jy*}) \Pr(x_j^* | x_j, y_j, \mathbf{c}_{jx*}) \Pr(y_j | x_j, \mathbf{c}_{jy}) \Pr(x_j | \mathbf{c}_{jx}) \quad (15)$$

The major distinction of Eqs. (14) and (15) from Eqs. (7) and (8) is the incorporation of differential misclassification via the modeling of the SE and SP parameters based on (12) and (13). Nondifferential misclassification may be viewed as a special case, corresponding to the assumption that $(\theta_1 = \dots = \theta_R = \theta_{R+2} = 0)$ in (12) or that $(\delta_1 = \dots = \delta_S = \delta_{S+2} = 0)$ in (13). The Akaike information criterion (AIC) [31] approach based on the internal validation study design offers the distinct advantage of helping to assess whether nondifferentiality best describes the data, as well as aiding in model selection for screening out factors associated with SE and SP for both X and Y . The scenario considered in this section is a generalization of that considered in [26], by allowing covariates in the main model and in the models for misclassification.

2.2.3. Dependent and differential misclassification. In the previous section, the diagnostic properties (SE and SP) of X and Y were allowed to be impacted by the true values of other variables, allowing flexible models for differential misclassification. With two binary variables subject to error, one must also recognize the potential for ‘dependent’ misclassification (e.g., [10]), where the SE and SP for one variable may be conditionally dependent on the error-prone assessment of the other. To clarify this notion of dependence in our setting, consider the following main study likelihood contribution:

$$\begin{aligned} \Pr(Y^* = y^*, X^* = x^* | \mathbf{C} = \mathbf{c}) &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(y^*, x^*, y, x | \mathbf{c}) \\ &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(y^* | y, x, x^*, \mathbf{c}) \Pr(x^* | x, y, \mathbf{c}) \Pr(y | x, \mathbf{c}) \Pr(x | \mathbf{c}) \end{aligned} \quad (16)$$

The expression in (16) reflects a model that generalizes (6) and (11) to encompass both dependence and differentiality. The first term in the last line of Equation (16) reflects dependent misclassification by conditioning on X^* , that is, generalizing the SE and SP for Y^* to depend on X^* conditionally on $(Y, X, \text{ and } \mathbf{C})$. The full main/internal validation study likelihood under such a model remains $L = L_m \times L_v$, where now the likelihood contribution for a main study subject is

$$\sum_{y_i=0}^{y_i=1} \sum_{x_i=0}^{x_i=1} \Pr(y_i^* | y_i, x_i, x_i^*, c_{iy*}) \Pr(x_i^* | x_i, y_i, c_{ix*}) \Pr(y_i | x_i, c_{iy}) \Pr(x_i | c_{ix})$$

Each internal validation subject now makes the following likelihood contribution:

$$\Pr(y_j^* | y_j, x_j, x_j^*, c_{jy*}) \Pr(x_j^* | x_j, y_j, c_{jx*}) \Pr(y_j | x_j, c_{jy}) \Pr(x_j | c_{jx})$$

As before, proper model selection may make it possible to reduce the model to a simpler form. For example, one can rely on AIC to assess the appropriateness of independent misclassification in (16), using the following generalized logistic regression model for the error-prone binary Y^* :

$$\text{logit}[\Pr(Y^* = 1 | Y, X, X^*, C_1, C_2, \dots, C_r)] = \theta_0 + \sum_{r=1}^R \theta_r C_r + \theta_{R+1} Y + \theta_{R+2} X + \theta_{R+3} X^* \quad (17)$$

The AIC can be used to address whether to include θ_{R+3} .

2.2.4. Other types of misclassification. It should be noted that there are other possible misclassification mechanisms besides those discussed in the previous sections, which may be regarded as special cases of the model presented in Section 2.2.3. For example, in practice, both exposure and disease status could be nondifferentially misclassified, yet misclassification could be dependent [32]. In such a case, one may make the appropriate simple adjustments to the form of the likelihood. Other mechanisms include the possibility that disease and exposure status are independently misclassified, with the misclassification being differential for one and nondifferential for the other. All likelihood functions presented herein for the main/internal validation study design potentially permit convenient numerical maximization via general optimization facilities available in standard statistical software such as SAS NLMIXED (SAS Institute, Cary, NC, USA). Details about the optimization techniques available in NLMIXED can be found in SAS documentation [33]. Standard errors for estimates of secondary parameters (e.g., SE and SP) can be estimated via the delta method, which is also part of the routine procedure in NLMIXED. A readily adaptable SAS code for this purpose is attached in the Appendix, using the HERS data in Section 3 as an example. In practice, thorough model selection should be performed for the SE/SP models for X and Y using the joint main/validation sample, to assess whether dependence and/or differentiability is involved in the misclassification process.

2.3. Factorizing the general misclassification model

Note the following alternative factorization for the general likelihood contribution in (16):

$$\begin{aligned} \Pr(Y^* = y^*, X^* = x^* | C = c) &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(y^*, x^*, y, x | c) \\ &= \sum_{y=0}^{y=1} \sum_{x=0}^{x=1} \Pr(x^* | x, y, y^*, c) \Pr(y^* | y, x, c) \Pr(y | x, c) \Pr(x | c) \end{aligned}$$

The last two terms in the preceding final expression are identical to those in Eq. (16), although the misclassification models for X and Y are altered in the dependent and differential misclassification case. However, this alternative factorization has no effect on the specifications in Section 2.2 under less than fully general misclassification patterns. In practice, one can use AIC to select the best-fitting factorization for the misclassification mechanism in the general case.

2.4. Variable selection for $X|C$ and misclassification models

In principle, the covariates in the $X|C$ or misclassification models could differ from those in the primary response model of interest. However, caution should be applied in this regard. Consider a situation in which a variable, say V , is included in the $X|C$ model and Y is not independent on V conditional on X and

all other covariates. In that case, failing to include V in the response model will generally induce bias in the estimation. A similar argument holds for variables involved in misclassification models. All variables in misclassification models should be accounted for in the response model, unless there is evidence suggesting that it is legitimate to assume they are independent of Y conditional on X and other covariates in the response model. In Section 4.2, we have provided empirical evidence for the preceding argument. In practice, we make the following suggestion. First, the response model of interest can be selected based upon the scientific research question and preliminary model selection to screen out the maximal C vector for all models, including the $X|C$ and misclassification models. In other words, only a subset of C will be considered in the $X|C$ and misclassification models. Then an AIC-based model selection strategy can be conducted by fitting candidate models and choosing the one with the smallest AIC.

3. HIV Epidemiology Research Study example analysis

According to Eq. (1), age, race, risk cohort, and HIV status are considered candidate C covariates for all models. In order to demonstrate the performance of the proposed approach, we randomly selected one-fourth of the total HERS sample size at visit 4 ($n_v = 214$) into an internal validation subsample. Model selection on all 214 participants suggested a version of the $X|C$ model as follows:

$$\text{logit}[\Pr(\text{CULTURE TRICH} = 1)] = \gamma_0 + \gamma_1 \text{RACE} \quad (18)$$

where race is associated with TRICH status based on the gold standard culture method. Predictor selection based on those 214 women further suggested dependent and differential misclassification of the CLIN BV and WET TRICH methods. The selected SE/SP models for CLIN BV and WET TRICH are as follows:

$$\begin{aligned} \text{logit}[\Pr(\text{CLIN BV} = 1)] = & \theta_0 + \theta_1 \text{LAB BV} + \theta_2 \text{WET TRICH} + \theta_3 \text{CULTURE TRICH} \\ & + \theta_4 \text{RISKCHRT} + \theta_5 \text{HIVPOS} \end{aligned} \quad (19)$$

$$\text{logit}[\Pr(\text{WET TRICH} = 1)] = \delta_0 + \delta_1 \text{CULTURE TRICH} + \delta_2 \text{LAB BV} + \delta_3 \text{RISKCHRT} \quad (20)$$

Thus, the classification rates of CLIN BV are found to be dependent on the risk factors of HIV risk cohort and HIV status, implying differential misclassification. Conditional on these variables, the misclassification process for BV depends on the error-prone wet mount version of the TRICH diagnosis along with the culture TRICH diagnosis, implying dependent misclassification. Similarly, model (20) indicates that HIV risk cohort significantly impacts the SE and SP of the wet mount method, a typical example of differential misclassification. As a practical matter, it is useful to perform model selection on the internal validation sample for Eqs. (18)–(20) via a traditional likelihood ratio testing approach with a conservative significance threshold such as 0.1 or 0.15, to ensure important factors are not left out. As recommended in Section 2.4, AIC can then be applied to the joint model to compare candidate misclassification mechanisms.

The first model fitted in Table II reflects the joint modeling of Eqs. (1) and (18)–(20) via the likelihood contributions given in (16) for dependent and differential misclassification. Note that this analysis incorporating gold standard test results (Y, X) for only a random subsample of women yields the same interpretation as the ideal analysis (Table I), with all primary parameter estimates having similar magnitudes and the same directionalities. In contrast, results assuming independent differential or non-differential misclassification are much more similar to the results of the naïve analysis. For instance, when assuming independence but allowing differential misclassification, the estimate for TRICH status increases, with a similar magnitude as observed in the naïve analysis. If assuming independent and non-differential misclassification, a highly elevated estimate for HIV risk cohort and a negative estimated $\ln(\text{OR})$ for HIV status are obtained, in addition to the elevated estimate for TRICH status. Note that AIC favors the most general misclassification model, suggesting a need to account for both dependent and differential misclassification in the HERS example. This important observation clearly highlights the value of internal validation sampling for evaluating and modeling potentially complex misclassification mechanisms. We also note that the alternative factorization (Section 2.3) of the dependent differential misclassification model was assessed but was not selected because of an AIC value larger than that for the factorization in Table II.

Table II. Results of ML analysis of main/internal validation study data on 904 women at the fourth HIV Epidemiology Research Study visit ($n_m = 690$, $n_v = 214$).

Variable	$\hat{\beta}$ (SE)	$\widehat{OR}$ (95% CI)
Assuming dependent and differential misclassification ^a (AIC = 1771.5)		
Trichomoniasis (+ versus −)	0.64 (0.41)	1.90 (0.38, 3.42)
Age (years)	−0.05 (0.02)	0.95 (0.92, 0.98)
Race (Black versus others)	0.79 (0.24)	2.19 (1.16, 3.23)
HIV risk cohort (IDU versus sexual)	0.28 (0.26)	1.32 (0.64, 2.00)
HIV status (+ versus −)	0.23 (0.27)	1.26 (0.58, 1.94)
Assuming independent and differential misclassification ^b (AIC = 1777.7)		
Trichomoniasis (+ versus −)	1.06 (0.38)	2.88 (0.73, 5.02)
Age (years)	−0.05 (0.02)	0.95 (0.92, 0.98)
Race (Black versus others)	0.73 (0.24)	2.07 (1.09, 3.05)
HIV risk cohort (IDU versus sexual)	0.30 (0.26)	1.35 (0.65, 2.05)
HIV status (+ versus −)	0.21 (0.28)	1.23 (0.56, 1.90)
Assuming nondifferential misclassification ^c (AIC = 1809.8)		
Trichomoniasis (+ versus −)	1.38 (0.37)	3.99 (1.09, 6.90)
Age (years)	−0.05 (0.02)	0.95 (0.92, 0.99)
Race (Black versus others)	0.73 (0.24)	2.08 (1.11, 3.04)
HIV risk cohort (IDU versus sexual)	0.85 (0.22)	2.35 (1.32, 3.37)
HIV status (+ versus −)	−0.14 (0.23)	0.87 (0.47, 1.26)

IDU, intravenous drug use; ML, maximum likelihood; SE, standard error.

^aML estimates of primary parameters are obtained by jointly modeling Eqs. (1) and (18)–(20).

^bWET TRICH is removed from Eq. (19) to indicate independence. The assumption is not supported by the data from the AIC value.

^cNo covariates affect *SE* and *SP* of *Y* and *X* in Eqs. (19) and (20). The assumption is not supported by the data from the AIC value.

Table III. Results of ML analysis of main/internal validation study data on 904 women at the fourth HIV Epidemiology Research Study visit ($n_m = 690$, $n_v = 214$): estimates of Eq. (19) describing *SE/SP* of *Y** (*CLIN BV*).

ML estimates of coefficients in Eq. (19)		
Variable	$\hat{\theta}$ (95% CI)	<i>p</i> -value
<i>LAB BV</i> (+ versus −)	2.92 (1.99, 3.85)	<0.0001
Wet mount Trichomoniasis (+ versus −)	1.34 (0.33, 2.34)	0.01
Culture Trichomoniasis (+ versus −)	0.36 (−0.21, 0.94)	0.22
HIV risk cohort (IDU versus sexual)	0.91 (0.41, 1.41)	0.0004
HIV status (+ versus −)	−0.61 (−1.15, −0.06)	0.03

ML estimates of parameters in Eq. (19) are obtained by jointly modeling Eqs. (1) and (18)–(20).

IDU, intravenous drug use; *LAB BV*, laboratory-based bacterial vaginosis; ML, maximum likelihood.

Tables III and IV summarize the ML estimates for misclassification parameters characterizing *CLIN BV* and wet mount TRICH in different strata, based on the selected models in (19) and (20). Note that TRICH diagnosis, HIV risk cohort, and HIV status all significantly affect the diagnostic properties of *CLIN BV*. By holding other covariates constant, *SE* tends to be higher in wet mount or culture TRICH-positive patients and intravenous drug users, while lower in HIV-positive women. An opposite trend is observed for the *SP* estimates. For wet mount TRICH, intravenous drug users or culture TRICH-positive participants appear to have a greater *SE* than those at risk via sexual contact, while the test is highly specific in both groups.

Table IV. Results of ML analysis of main/internal validation study data on 904 women at the fourth HIV Epidemiology Research Study visit ($n_m = 690, n_v = 214$): estimates of Eq. (20) describing SE/SP of X^* (WET TRICH).

ML estimates of coefficients in Eq. (20)		
Variable	$\hat{\delta}$ (95% CI)	p -value
Culture Trichomoniasis (+ versus –)	4.13 (2.39, 5.87)	<0.0001
LAB BV (+ versus –)	0.45 (–0.26, 1.16)	0.21
HIV risk cohort (IDU versus sexual)	1.28 (0.55, 2.00)	0.001

ML estimates of parameters in Eq. (20) are obtained by jointly modeling Eqs. (1) and (18)–(20).

IDU, intravenous drug use; LAB BV, laboratory-based bacterial vaginosis;

ML, maximum likelihood.

One might wonder whether fitting the full joint likelihood using the complete set of error-prone and gold standard test results for BV and TRICH would yield different results than those based only on the gold standard data. The wide availability of the four variables (X , Y , X^* , and Y^*) in the HERS permits this unique comparison, and we find unsurprisingly that the results are exactly the same as for the ‘ideal analysis’ in Table I. The reason is that with (X , Y , X^* , Y^*) available on all subjects, the likelihood factors into four separate models (Eqs. (1) and (18)–(20)). In that case, estimates of the parameters in Eq. (1) do not rely on estimates from the other models.

4. Simulation studies

4.1. Data-driven simulations

The simulation experiment summarized in Table V demonstrates the empirical performance of estimation based on the joint main/internal validation likelihood, under dependent differential misclassification conditions mimicking those observed at HERS visit 4. Specifically, data were generated jointly under the primary model of interest (1) and the logistic models in (18)–(20), with true coefficients and covariate distributions similar to those estimated and observed in the HERS example. The ideal and naïve analyses (as in Table I) were based on 500 simulated replicates of (X , Y) and (X^* , Y^*), respectively, while the complete analysis was conducted via joint ML using the main and internal validation data corresponding to each replication as described in Section 2. The internal validation subsample was randomly selected from 25%, 10%, and 5% of the total sample, while the total sample size was 904, as in the HERS example.

Table V confirms that the naïve analysis yields markedly biased estimates, with correspondingly poor CI coverage. The joint likelihood-based analysis assuming dependent and differential misclassification, however, produces reliable results with mean coefficient estimates very close to the true values and excellent 95% CI coverage, given a sufficient internal validation sample. Empirical evidence in this setting mimicking the HERS suggests that an internal validation sample of approximately 15% of the total sample size (i.e., 226) is sufficient to ensure valid parameter estimates corrected for complex misclassification. As the internal sample size drops to 10% (around 90 subjects) or 5% (around 45 subjects), the corrected estimate of the exposure coefficient in particular begins to suffer bias predictably, and inflated standard errors are also observed with smaller internal validation samples. In practice, we thus recommend empirical examinations at the study planning stage based upon a series of assumptions about the misclassification processes. Simulation studies using programs available from the lead author can facilitate such examinations to inform the necessary total sample size and validation fraction.

4.2. Impact of $X|C$ model misspecification

Simulation studies were also conducted to examine the impact of misspecifying the $X|C$ model on estimating primary parameters. In hypothetical examples, data were simulated based on Eqs. (1) and

Table V. Results of simulations designed to mimic conditions of HIV Epidemiology Research Study example.

Variable	$\hat{\beta}$ (SE) ^a	95% CI coverage %
Ideal analysis ^b		
Trichomoniasis (+ versus −)	0.65 (0.25)	96.6
Age (years)	−0.05 (0.003)	93.2
Race (Black versus others)	0.80 (0.19)	96.4
HIV risk cohort (IDU versus sexual)	0.28 (0.18)	95.0
HIV status (+ versus −)	0.25 (0.19)	96.4
Naïve analysis ^c		
Trichomoniasis (+ versus −)	1.54 (0.29)	12.2
Age (years)	−0.02 (0.002)	0
Race (Black versus others)	0.38 (0.18)	37.2
HIV risk cohort (IDU versus sexual)	0.83 (0.18)	12.8
HIV status (+ versus −)	−0.42 (0.18)	5.8
Complete analysis ($n_v = 25\% \times n$) ^d		
Trichomoniasis (+ versus −)	0.61 (0.49)	94.6
Age (years)	−0.05 (0.01)	96.0
Race (Black versus others)	0.82 (0.34)	95.0
HIV risk cohort (IDU versus sexual)	0.28 (0.33)	94.6
HIV status (+ versus −)	0.23 (0.35)	95.6
Complete analysis ($n_v = 15\% \times n$) ^e		
Trichomoniasis (+ versus −)	0.58 (0.62)	94.2
Age (years)	−0.05 (0.01)	93.6
Race (Black versus others)	0.82 (0.40)	94.2
HIV risk cohort (IDU versus sexual)	0.28 (0.36)	95.0
HIV status (+ versus −)	0.24 (0.42)	95.8
Complete analysis ($n_v = 10\% \times n$) ^f		
Trichomoniasis (+ versus −)	0.50 (0.76)	93.0
Age (years)	−0.05 (0.01)	93.6
Race (Black versus others)	0.84 (0.45)	94.0
HIV risk cohort (IDU versus sexual)	0.28 (0.48)	93.6
HIV status (+ versus −)	0.26 (0.50)	93.6
Complete analysis ($n_v = 5\% \times n$) ^g		
Trichomoniasis (+ versus −)	0.23 (1.05)	84.6
Age (years)	−0.05 (0.02)	93.0
Race (Black versus others)	0.94 (0.58)	94.4
HIV risk cohort (IDU versus sexual)	0.35 (0.62)	95.2
HIV status (+ versus −)	0.29 (0.65)	93.4

Five hundred simulations. $n_m = 690$, $n_v = 214$. Data were generated from Eqs. (1) and (18)–(20). True parameters: ($\beta_0 = 0.63$, $\beta_1 = 0.64$, $\beta_2 = -0.05$, $\beta_3 = 0.79$, $\beta_4 = 0.28$, $\beta_5 = 0.23$, $\theta_0 = -3.27$, $\theta_1 = 2.92$, $\theta_2 = 1.34$, $\theta_3 = 0.36$, $\theta_4 = 0.91$, $\theta_5 = -0.61$, $\gamma_0 = -3.43$, $\gamma_1 = 2.48$, $\delta_0 = -5.64$, $\delta_1 = 4.13$, $\delta_2 = 0.45$, $\delta_3 = 1.28$).

IDU, intravenous drug use; ML, maximum likelihood; SE, standard error.

^aEmpirical standard deviations across 500 estimates are reported in parentheses.

^bML estimates from Eq. (1).

^cML estimates from Eq. (1) with (Y^* , X^*) replacing (Y , X).

^dML estimates of primary parameters are obtained by jointly modeling Eqs. (1) and (18)–(20). The internal validation sample accounts for 25% of the total sample; 500/500 replicates converged.

^eThe internal validation sample accounts for 15% of the total sample; 499/500 replicates converged.

^fThe internal validation sample accounts for 10% of the total sample; 497/500 replicates converged.

^gThe internal validation sample accounts for 5% of the total sample; 485/500 replicates converged.

Table VI. Results of examining impact of misspecifying $X|C$ model when HIV status is important in $X|C$ and conditionally dependent on response.

Variable	$\hat{\beta}$ (SE) ^a	95% CI coverage (%)
Correct model ^b		
Trichomoniasis (+ versus −)	0.60 (0.44)	
Age (years)	−0.05 (0.01)	
Race (Black versus others)	0.82 (0.35)	
HIV risk cohort (IDU versus sexual)	0.28 (0.32)	
HIV status (+ versus −)	1.05 (0.33)	
$X C$ is misspecified ^c		
Trichomoniasis (+ versus −)	0.58 (0.44)	
Age (years)	−0.05 (0.01)	
Race (Black versus others)	0.82 (0.33)	
HIV risk cohort (IDU versus sexual)	0.27 (0.32)	
HIV status (+ versus −)	1.09 (0.33)	
Primary model is misspecified ^d		
Trichomoniasis (+ versus −)	0.88 (0.45)	
Age (years)	−0.05 (0.01)	
Race (Black versus others)	0.65 (0.32)	
HIV risk cohort (IDU versus sexual)	0.26 (0.33)	
HIV status (+ versus −)	N/A	
ML estimates from AIC-based selected model ^e		
Trichomoniasis (+ versus −)	0.60 (0.44)	94.8
Age (years)	−0.05 (0.01)	95.2
Race (Black versus others)	0.82 (0.34)	96.8
HIV risk cohort (IDU versus sexual)	0.28 (0.32)	94.0
HIV status (+ versus −)	1.07 (0.33)	95.2

Five hundred simulations. $n_m = 690$, $n_v = 214$. Data were simulated from Eqs. (1) and (18)–(20). True parameters: $(\beta_0 = 0.63, \beta_1 = 0.64, \beta_2 = -0.05, \beta_3 = 0.79, \beta_4 = 0.28, \beta_5 = 1, \theta_0 = -3.27, \theta_1 = 2.92, \theta_2 = 1.34, \theta_3 = 0.36, \theta_4 = 0.91, \gamma_0 = -3.43, \gamma_1 = 2.48, \gamma_2 = 1$ for HIV status, $\delta_0 = -5.64, \delta_1 = 4.13, \delta_2 = 0.45, \delta_3 = 1.28)$.

IDU, intravenous drug use; ML, maximum likelihood; N/A, not applicable; SE, standard error.

^aEmpirical standard deviations across 500 estimates are reported in parentheses.

^bML estimates of primary parameters are obtained by jointly modeling Eqs. (1) and (18)–(20) with correct model specification.

^cML estimates of primary parameters are obtained by jointly modeling Eqs. (1) and (18)–(20) with Eq. (18) misspecified, in which HIV status was omitted.

^dML estimates of primary parameters are obtained by jointly modeling Eqs. (1) and (18)–(20) with Eq. (1) misspecified, in which HIV status was omitted.

^eThe model specifying both $X|C$ and primary models was correctly selected 96% of the time.

(18)–(20) with HIV status as an important predictor in both the primary $Y|X, C$ model and the $X|C$ model. A total of 500 simulations were performed in Table VI, which also provides the true parameter values and overall and internal validation sample sizes. Three candidate models were fit under each scenario: the correct model, a model misspecifying the $X|C$ model, and a model misspecifying the primary model.

Table VI shows that when HIV status is important in both the $X|C$ and primary models, failing to adjust for this factor in the $X|C$ model appears to slightly bias the estimate of the coefficient of HIV status in the primary model. When the $X|C$ model is correctly specified but HIV status is omitted from the primary model, the estimated coefficients for TRICH and race in the latter are biased. We also simulated the scenario when HIV status is only important in the $X|C$ model. In this case, we observe that the accuracy in estimating primary parameters remains (results not shown). Similar findings have been observed in simulations when misclassification models are subject to misspecification. Empirical observations support adopting the strategy recommended in Section 2.4, that is, that one should restrict the attention only

to covariates selected in the primary model when building the $X|C$ and misclassification models. Perhaps most importantly, we find that the AIC-based model selection strategy selected the correct model in most cases.

5. Discussion

In this manuscript, we have expanded upon previously demonstrated likelihood-based methodology for dealing with misclassification via a main/internal validation study design. The approach may be viewed as a generalization of aspects of prior work (e.g., [2, 26]) to incorporate validation data and covariates into both the main model of interest and misclassification models. It also generalizes the logistic regression-based approach outlined in [24], in which only outcome misclassification was addressed. The joint likelihood is developed for various combinations of differential and/or dependent misclassification of two binary variables (Y and X), yielding special case extensions of general main/internal validation study likelihood specifications described in [14]. Specific motivation for these developments is provided by cross-sectional data involving assessments of BV and TRICH status and covariates measured in the HERS.

Our goal has been to provide clear guidance on adjusting for biases due to misclassification in binary response and predictor variables in logistic regression, when resources permit the collection of a reasonably sized internal sample with validations of both variables. We strongly emphasize the importance of such internal validation sampling to assess misclassification patterns and to ensure the validity of the results, noting that only this study design-based tactic permitted the identification of the complex misclassification mechanisms operating in the HERS. The parametric approach taken here has the advantage of accessibility, with SAS NLMIXED programs provided in the Appendix. It also makes AIC calculations available to assist with model selection and mechanism evaluation, as demonstrated in the HERS example.

Although throughout we have adopted logit links for all models, other links can also be used without noteworthy additional conceptual or technical difficulty. Compared with alternative approaches such as sensitivity analyses, our approach along with the collection of internal validation data allows much greater flexibility when the misclassification process is complex, as evident in the HERS data with respect to BV and TRICH status. In contrast, sensitivity analyses may be preferred if there is a strong reason to believe that misclassification in both variables is nondifferential and if there is reasonably reliable preliminary knowledge regarding the operating characteristics (SE and SP) of the error-prone tests measuring the response and the exposure. However, if misclassification may be complex, it would be valuable to make efforts to collect validation information.

As with any parametric approach, model selection can be critical in our setting in order to ensure valid estimation. We have suggested a practical strategy in Section 2.4, in which candidate covariates in the $X|C$ and SE/SP models should only be selected from covariates in the primary model, unless one has confidence that additional covariates introduced into the $X|C$ or misclassification models are conditionally independent from the response. Otherwise, bias may occur in the estimation. An alternative analytic strategy could borrow ideas from multiple imputation [34], via which one could base imputation of the value of X on a model that includes other correctly measured covariates (C). Cole *et al.* [35] demonstrate the application of multiple imputation when X is misclassified. Although only nondifferential misclassification was discussed in their work, their approach could potentially be extended to more general situations that might include differential misclassification and the case of both X and Y being subject to misclassification. Compared with existing alternatives, we note that our ML approach has been generalized to complex misclassification in both exposure and response variables and is computationally accessible. It also allows for AIC-based model selection, which makes it possible to carefully study and account for the misclassification pattern. One should always note that, like in all model selection problems, the primary response model should be specified in light of the research question as well as scientific knowledge, in addition to statistical considerations.

In a separate work [36], we have studied methods to adjust for differential misclassification of BV status longitudinally within the HERS. Future work may include efforts to extend these regression-based correction approaches to adjust for both outcome and predictor misclassification when both are repeatedly measured over time.

Appendix

Sample SAS NLMIXED code applied to HERS example as described in Section 3.

```
PROC NLMIXED data=data;
parms  theta0=-2 theta1=2 theta2=1 theta3=0.5 theta4= 0.8 theta5=-0.6
delta0= -5 delta1=4 delta2=0.5 delta3=1 beta0= 0.1 beta1= 0.9 beta2=-
0.04 beta3=0.8 beta4=0.3 beta5=0.2 gamma0= -2 gamma1= 2.5;/*specify
initial values*/

eta1_y1=theta0+theta1+theta2*xstar+theta3*x+theta4*riskchrt+theta5*hiv
pos;/*misclassification model of Y*/
pa_y1=exp(eta1_y1)/(1+exp(eta1_y1));
eta1_y0= theta0+theta2*xstar+theta3*x+theta4*riskchrt+theta5*hivpos;
pa_y0=exp(eta1_y0)/(1+exp(eta1_y0));

eta2_1=delta0+delta1+delta2*y +delta3*riskchrt; /*misclassification
model of X*/
pb_1=exp(eta2_1)/(1+exp(eta2_1));
eta2_0= delta0+delta2*y +delta3*riskchrt;
pb_0=exp(eta2_0)/(1+exp(eta2_0));

eta3_x1=beta0+beta1+beta2*age+beta3*black+beta4*riskchrt+beta5*hivpos;
/*primary model*/
pc_x1=exp(eta3_x1)/(1+exp(eta3_x1));
eta3_x0=beta0+beta2*age+beta3*black+beta4*riskchrt+beta5*hivpos;
pc_x0=exp(eta3_x0)/(1+exp(eta3_x0));

eta4=gamma0+gamma1*black;/*X|C model*/
pd=exp(eta4)/(1+exp(eta4));

likel=((pa_y1)**ystar*(1-pa_y1)**(1-ystar)*(pb_1)**xstar*(1-pb_1)**(1-
xstar)*(pc_x1)*pd+
(pa_y0)**ystar*(1-pa_y0)**(1-ystar)*(pb_1)**xstar*(1-pb_1)**(1-
xstar)*(1-pc_x1)*pd +
(pa_y1)**ystar*(1-pa_y1)**(1-ystar)*(pb_0)**xstar*(1-pb_0)**(1-
xstar)*(pc_x0)*(1-pd) +
(pa_y0)**ystar*(1-pa_y0)**(1-ystar)*(pb_0)**xstar*(1-pb_0)**(1-
xstar)*(1-pc_x0)*(1-pd))**val;/*likelihood of main study*/

eta1=theta0+theta1*y+theta2*xstar+theta3*x+theta4*riskchrt+theta5*hivp
os;
pa=exp(eta1)/(1+exp(eta1));

eta2= delta0+delta1*x+delta2*y +delta3*riskchrt;
pb=exp(eta2)/(1+exp(eta2));

eta3= beta0+beta1*x+beta2*age+beta3*black+beta4*riskchrt+beta5*hivpos;
pc=exp(eta3)/(1+exp(eta3));

like2=(pa**ystar*(1-pa)**(1-ystar)*pb**xstar*(1-pb)**(1-
xstar)*pc**y*(1-pc)**(1-y)*pd**x*(1-pd)**(1-x))**val;/*likelihood of
validation subsample*/

like = likel*like2;

loglik=log(like);
model ystar ~ general(loglik);
run;
```

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